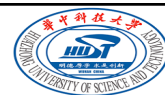


人工智能与自动化学院

模式识别与机器学习



第十一讲 卷积神经网络(*Convolutional Neural Networks*)



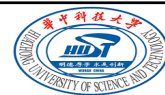
11.1 背景(*Background*)

11.2 卷积核的知识(*About Convolutional Kernels*)

11.3 卷积神经网络(*Convolutional Neural Networks*)

11.4 应用示例(*Case Study*)

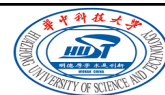
11.1 背景



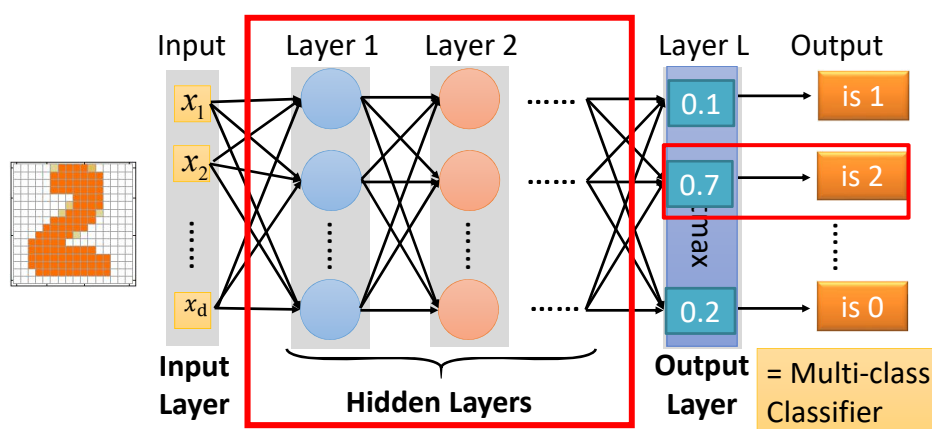
深度学习存在的挑战和关键技术

- 如何设计或确定网络结构
 - “领域知识”有助于选择网络结构，如计算机视觉中常常用卷积神经网络
- 模型复杂度高导致过拟合
 - “大数据”有助于增加训练集，提升泛化能力
 - “正则化”——如Dropout等手段，增强对噪声的容忍度
- 模型复杂难以找到最优解
 - 仔细选择初始值，避免落入局部极值，包括预训练等
 - 防止梯度消失，采用合适的激活函数、设计更好的网络结构
- 计算复杂度高
 - 并行化、批量化、GPU等

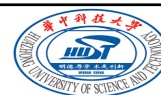
11.1 背景



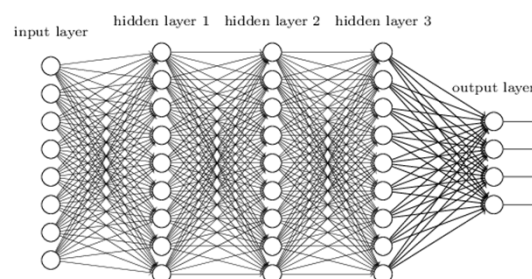
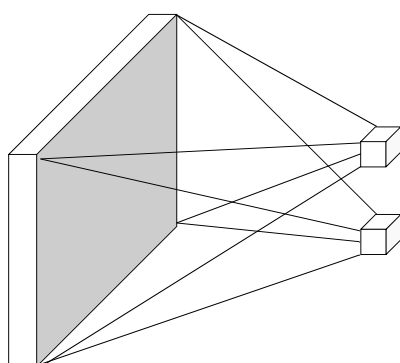
神经网络(Neural Networks):



11.1 背景



神经网络(Neural Networks):

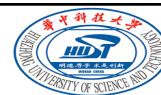


全连接层(Fully connected layer)

Image Size in CIFAR10: $32 \times 32 \times 3 = 3072$

Image Size : $200 \times 200 \times 3 = 120000$

第十一讲 卷积神经网络(Convolutional Neural Networks)



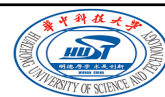
11.1 背景(Background)

11.2 卷积核的知识(About Convolutional Kernels)

11.3 卷积神经网络(Convolutional Neural Networks)

11.4 应用示例(Case Study)

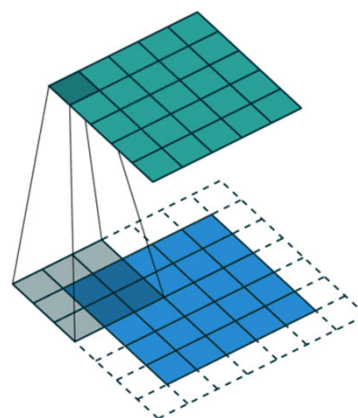
11.2 卷积核的知识



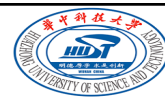
二维卷积(*convolution in 2D*):

$$f[n, m] * h[n, m]$$

$$= \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] h[n - k, m - l]$$



11.2 卷积核的知识



二维卷积应用示例(*convolution in 2D – examples*):



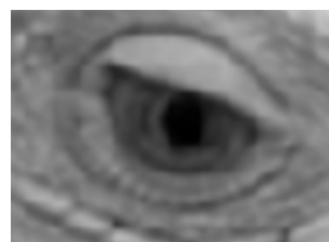
f[m, n]

$$* \frac{1}{9}$$

•1	•1	•1
•1	•1	•1
•1	•1	•1

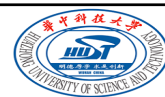
h[m, n]

=



g[m, n]

11.2 卷积核的知识



二维卷积应用示例(*convolution in 2D – examples*):



$f[m,n]$

$\otimes$

$[-1, 1]$

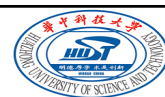
$=$

$h[m,n]$



$g[m,n]$

11.2 卷积核的知识



二维卷积应用示例(*convolution in 2D – examples*):



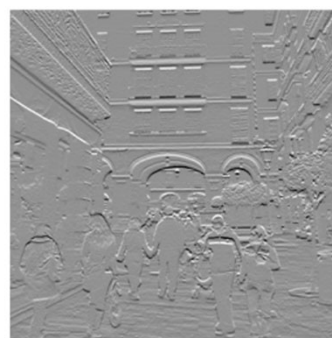
$f[m,n]$

$\otimes$

$[-1, 1]^T$

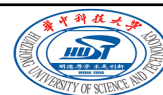
$=$

$h[m,n]$

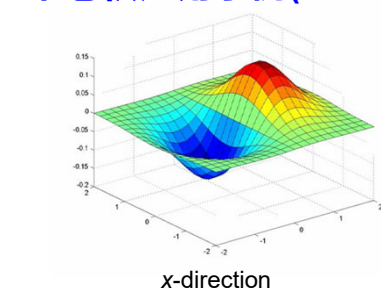


$g[m,n]$

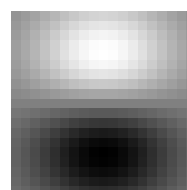
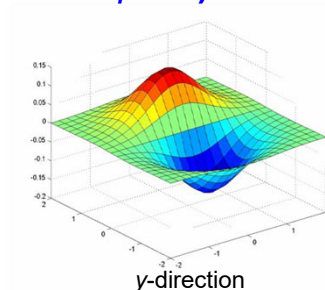
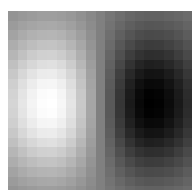
11.2 卷积核的知识



二维卷积应用示例(*convolution in 2D – examples*):

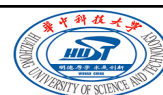


$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}$$

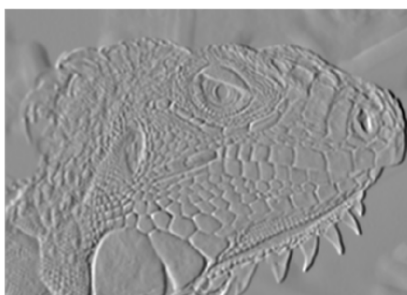


$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

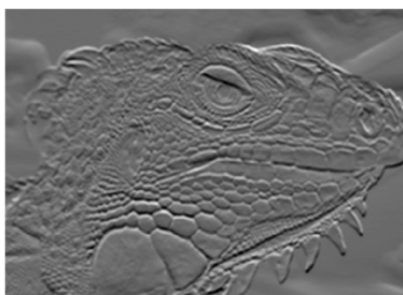
11.2 卷积核的知识



二维卷积应用示例(*convolution in 2D – examples*):

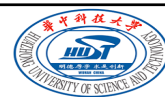


X-Derivative of Gaussian

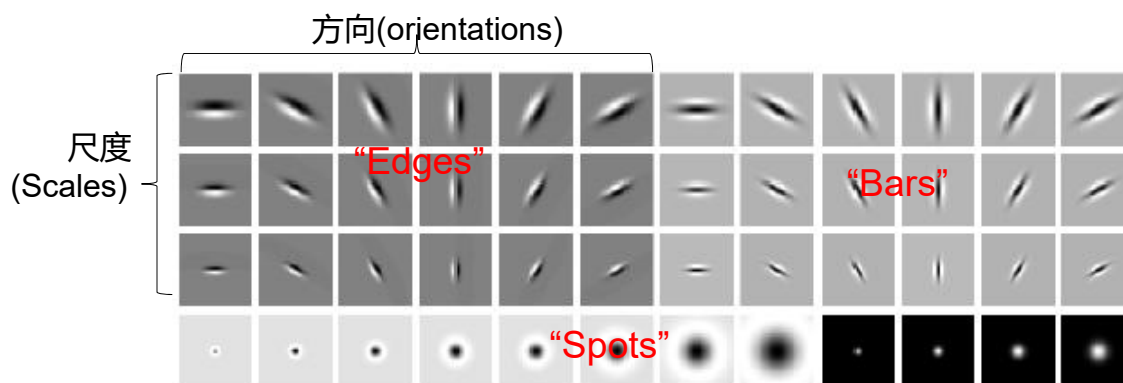


Y-Derivative of Gaussian

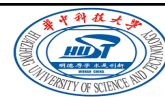
11.2 卷积核的知识



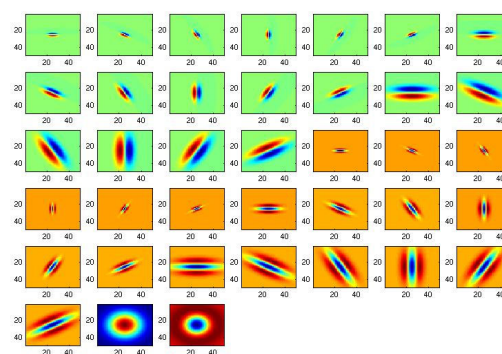
滤波器组(Filter banks):



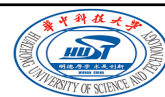
11.2 卷积核的知识



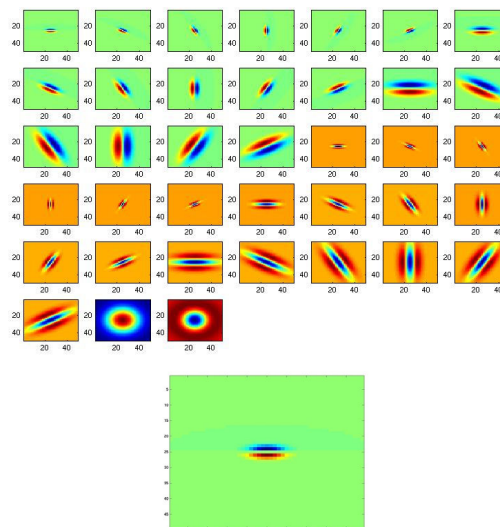
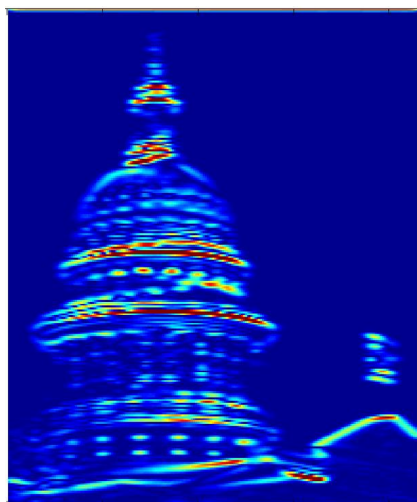
滤波器组(Filter banks):



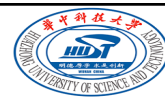
11.2 卷积核的知识



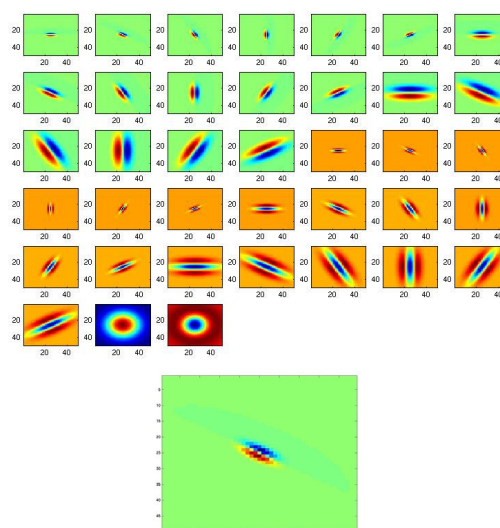
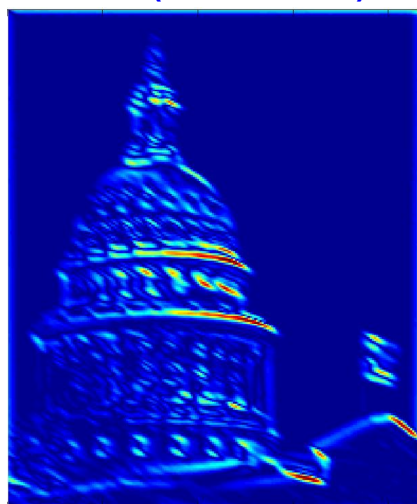
滤波器组(Filter banks):



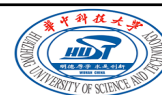
11.2 卷积核的知识



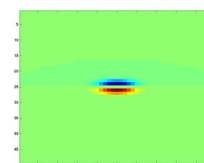
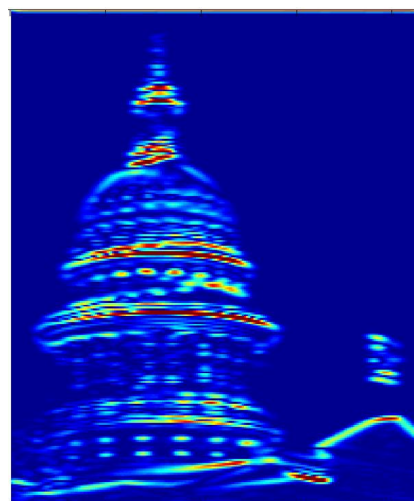
滤波器组(Filter banks):



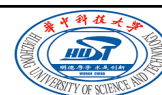
11.2 卷积核的知识



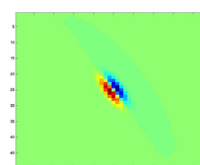
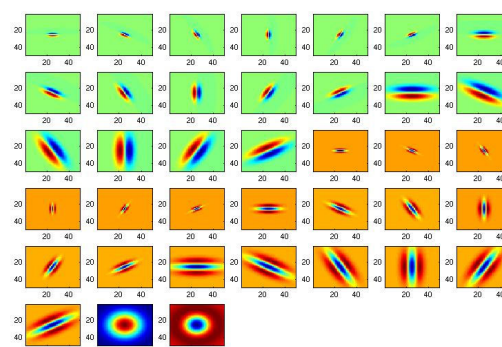
滤波器组(Filter banks):



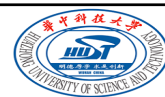
11.2 卷积核的知识



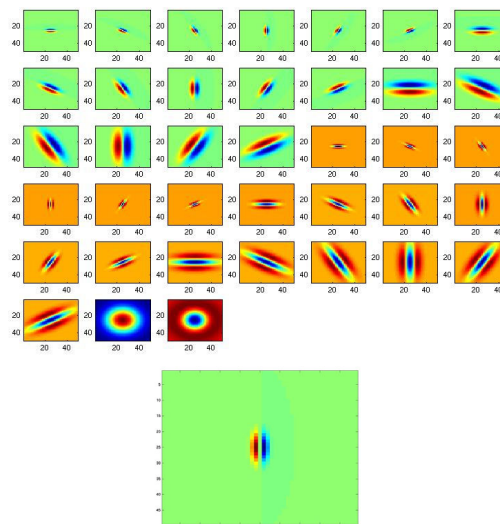
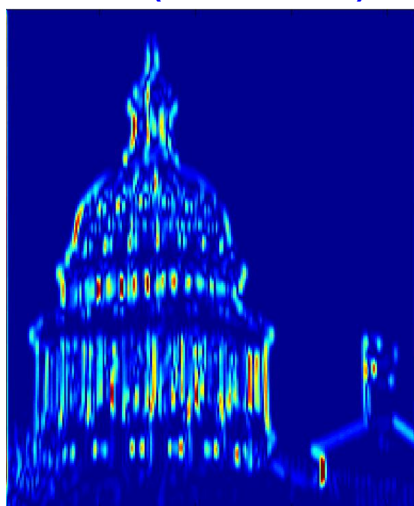
滤波器组(Filter banks):



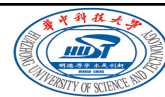
11.2 卷积核的知识



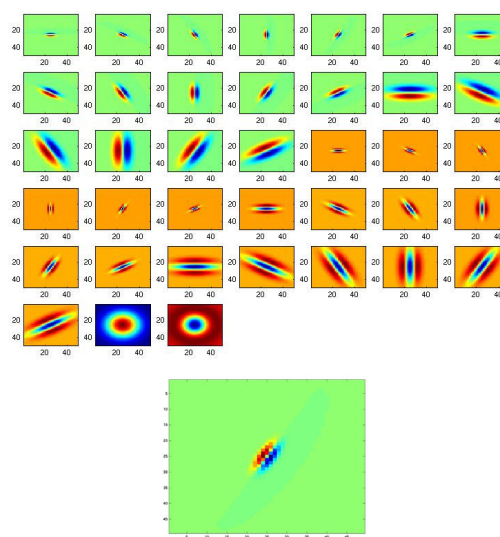
滤波器组(Filter banks):



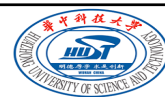
11.2 卷积核的知识



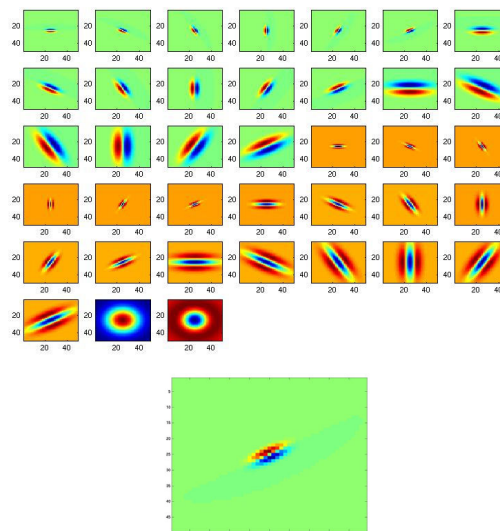
滤波器组(Filter banks):



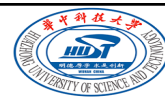
11.2 卷积核的知识



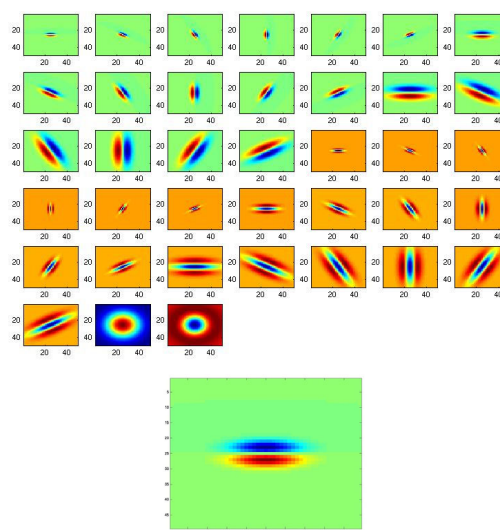
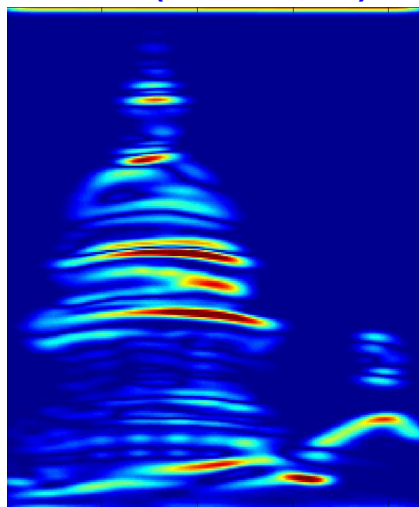
滤波器组(Filter banks):



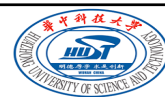
11.2 卷积核的知识



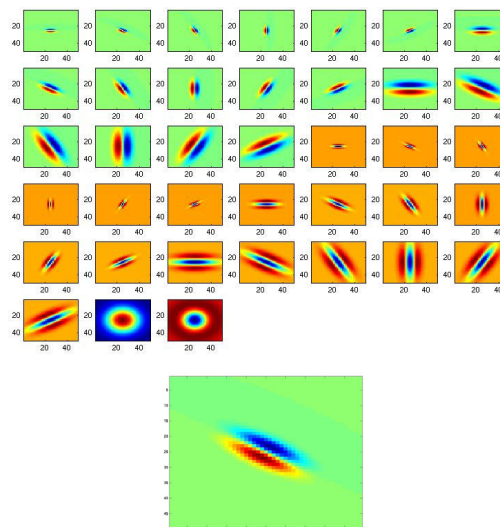
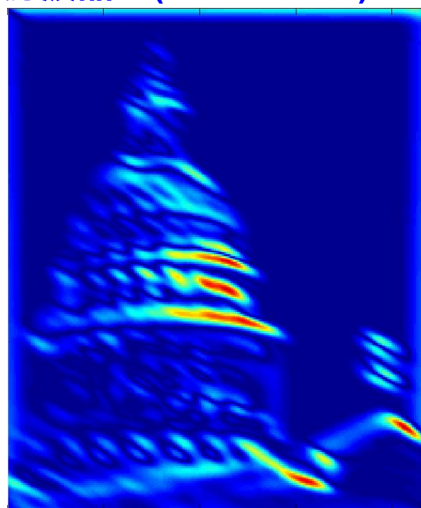
滤波器组(Filter banks):



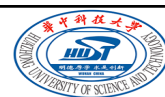
11.2 卷积核的知识



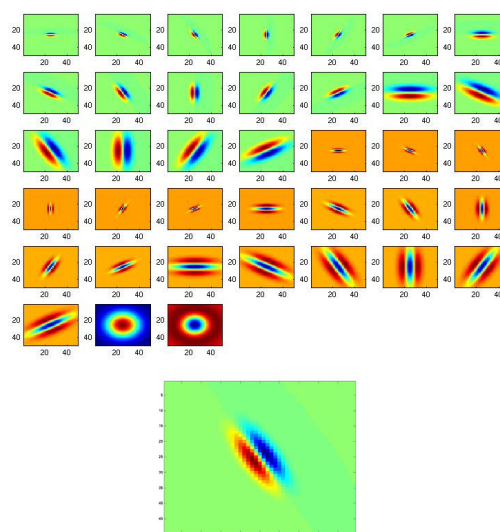
滤波器组(Filter banks):



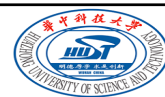
11.2 卷积核的知识



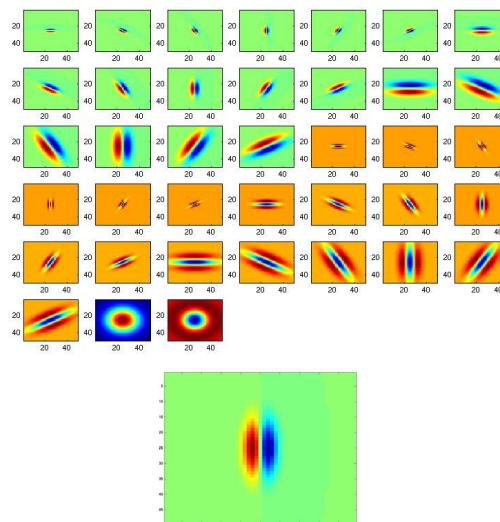
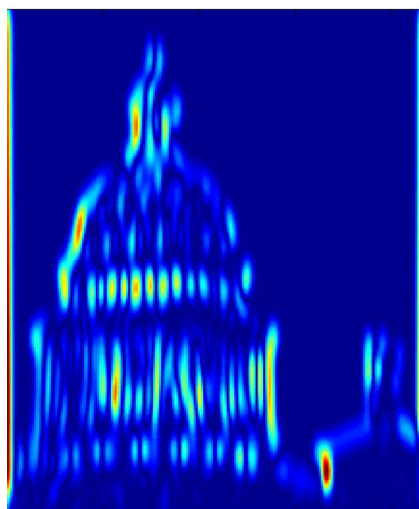
滤波器组(Filter banks):



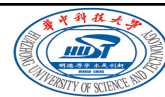
11.2 卷积核的知识



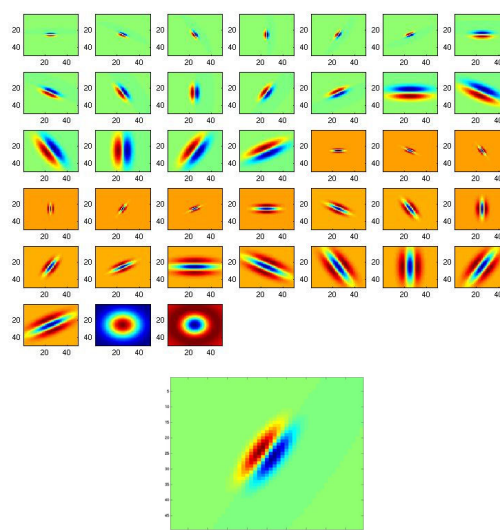
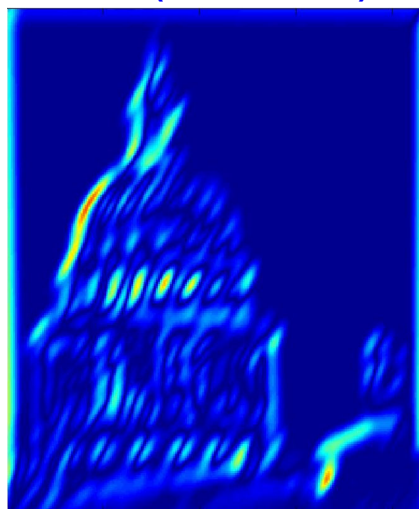
滤波器组(Filter banks):



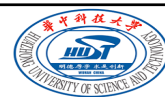
11.2 卷积核的知识



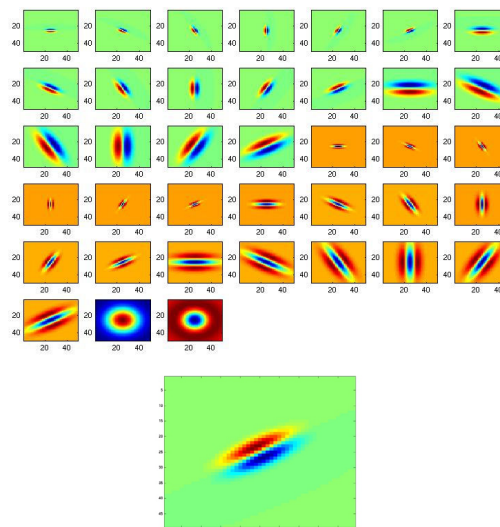
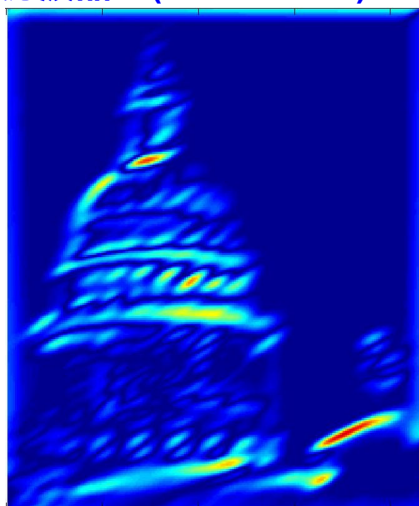
滤波器组(Filter banks):



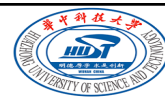
11.2 卷积核的知识



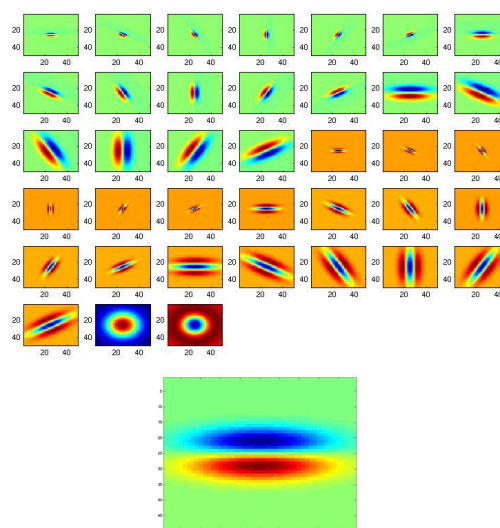
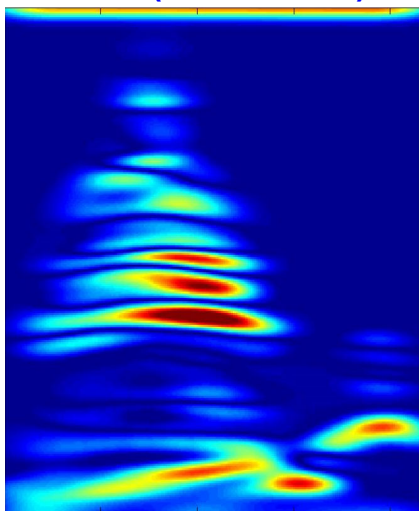
滤波器组(Filter banks):



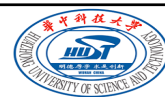
11.2 卷积核的知识



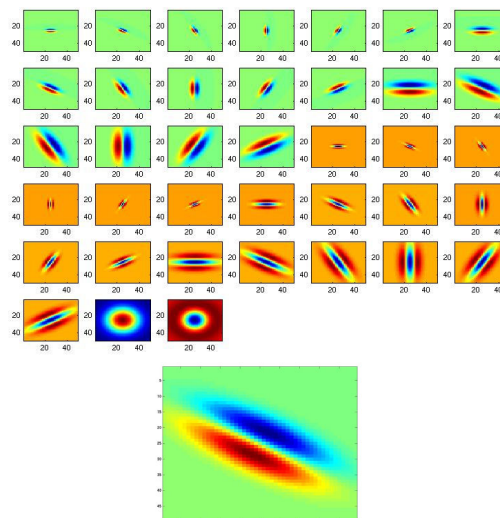
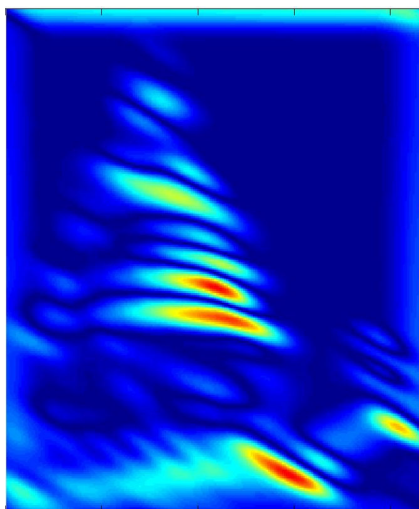
滤波器组(Filter banks):



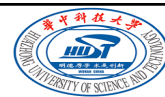
11.2 卷积核的知识



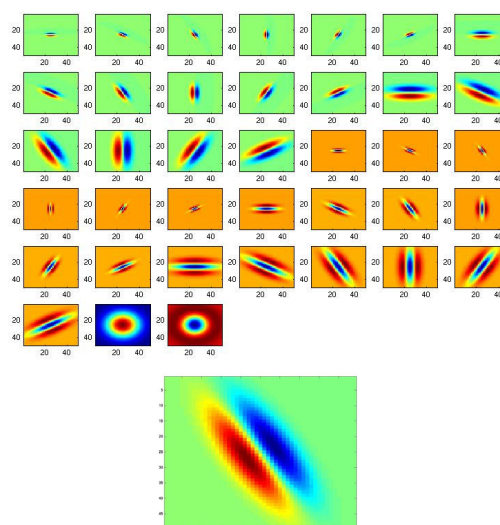
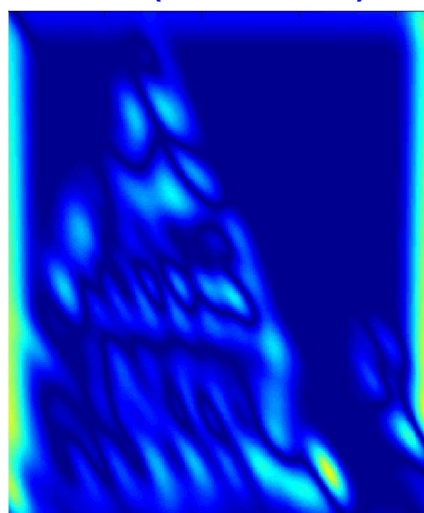
滤波器组(Filter banks):



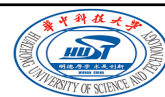
11.2 卷积核的知识



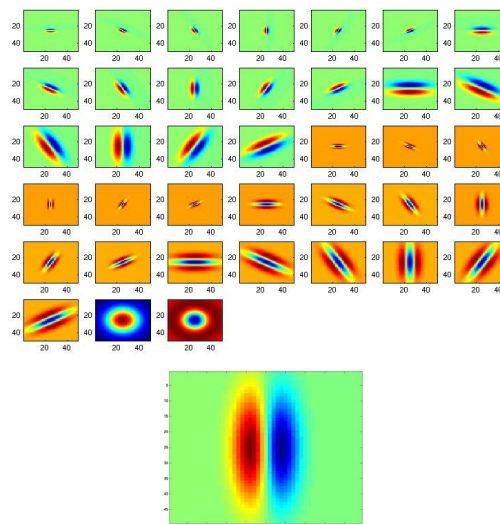
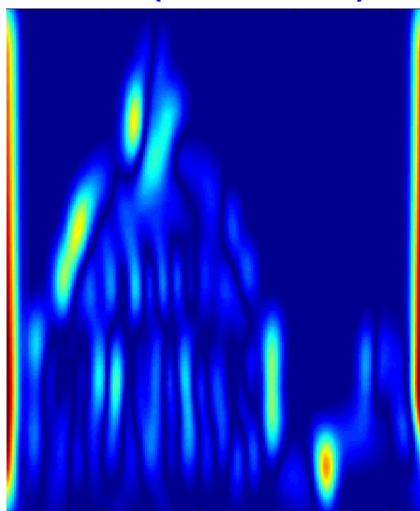
滤波器组(Filter banks):



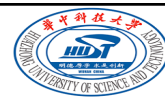
11.2 卷积核的知识



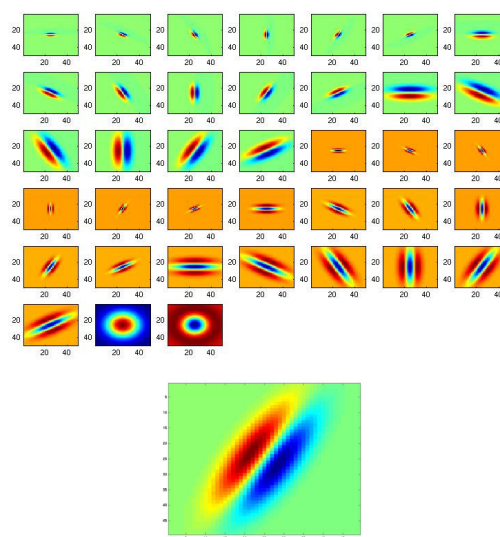
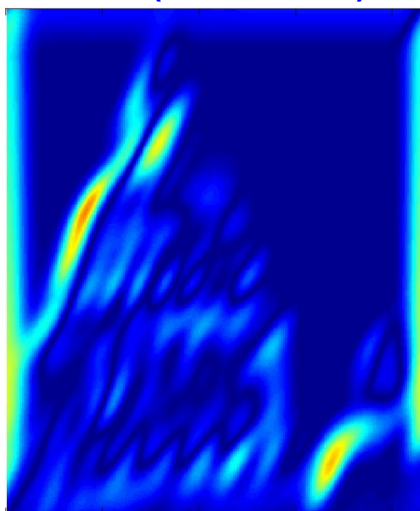
滤波器组(Filter banks):



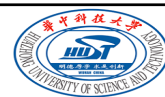
11.2 卷积核的知识



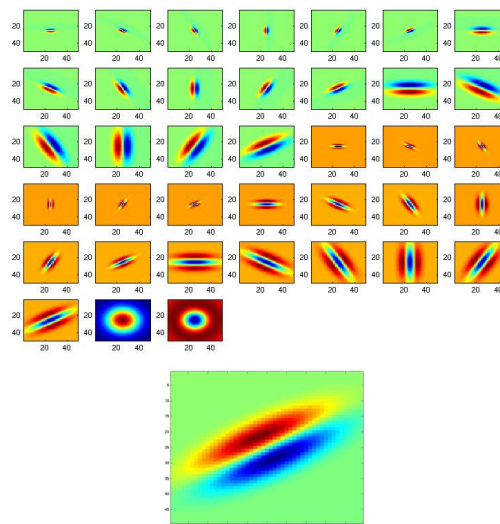
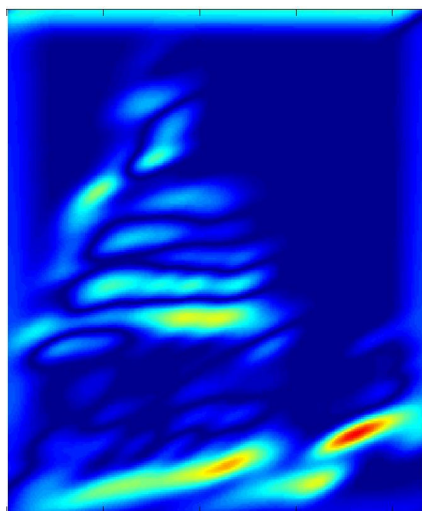
滤波器组(Filter banks):



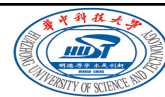
11.2 卷积核的知识



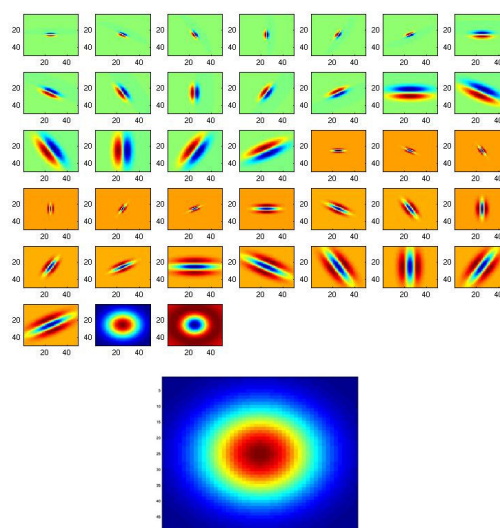
滤波器组(Filter banks):



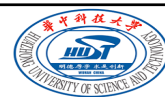
11.2 卷积核的知识



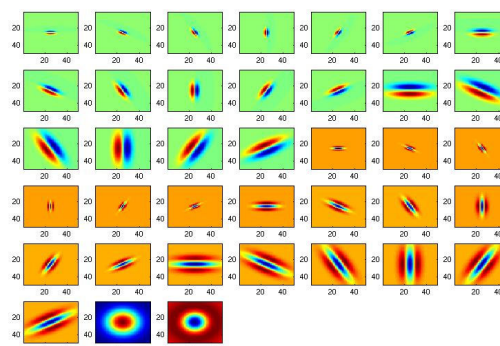
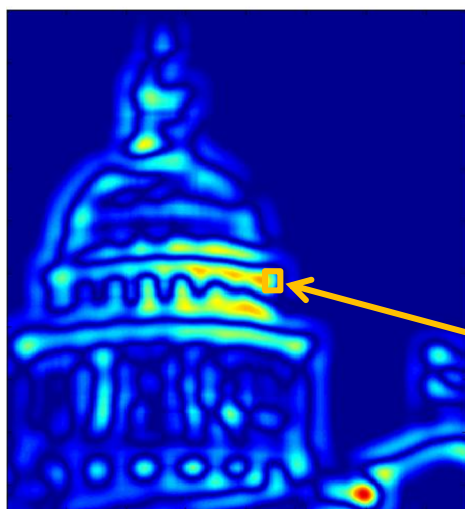
滤波器组(Filter banks):



11.2 卷积核的知识



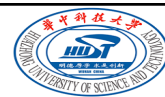
滤波器组(Filter banks):



[r1, r2, ..., r38]

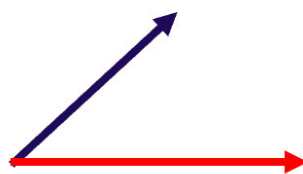
每个像素上由滤波器组产生
一系列响应构造成特征向量

2.1 感知器模型参数空间



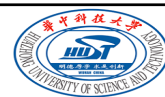
感知器模型的可视化:

$$f[n, m] * h[n, m] = \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] h[n - k, m - l]$$

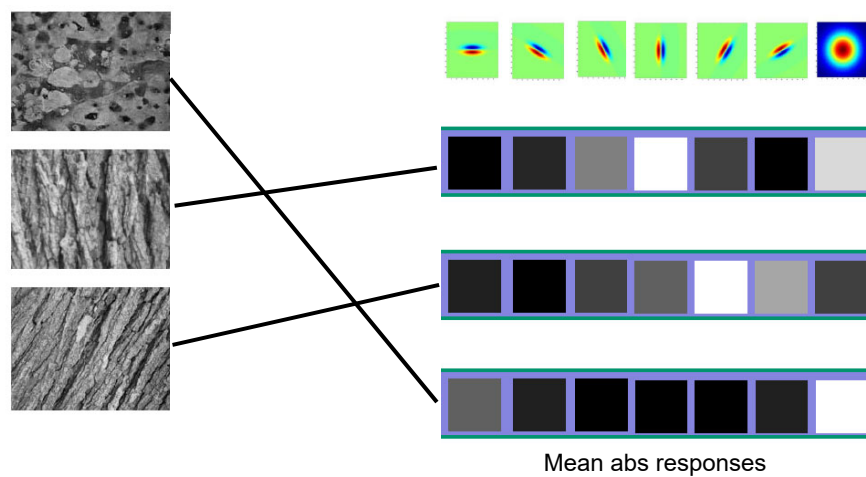


$$\mathbf{W}^T \mathbf{X} = \|\mathbf{W}\| \|\mathbf{X}\| \cos \theta$$

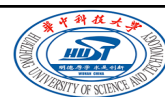
11.2 卷积核的知识



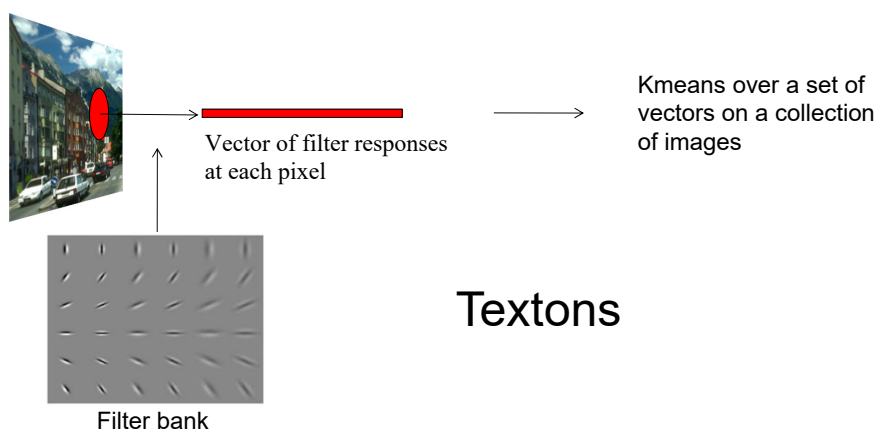
滤波器组(Filter banks):



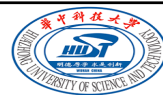
11.2 卷积核的知识



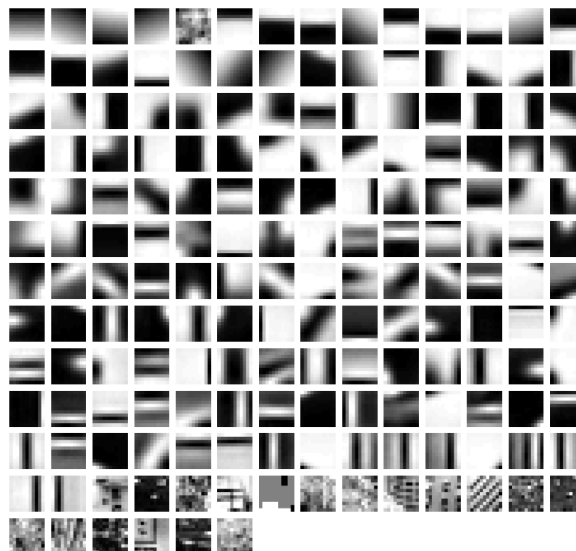
滤波器组(Filter banks):



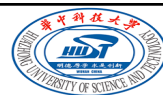
11.2 卷积核的知识



Codewords
dictionary
formation



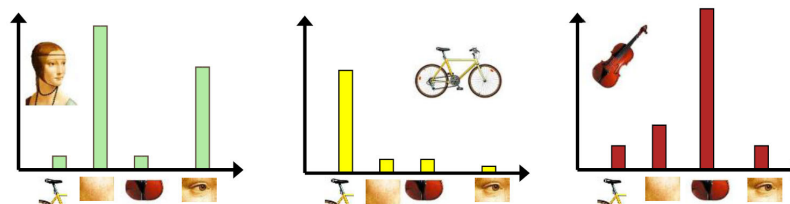
11.2 卷积核的知识



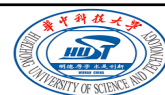
1. Extract features
2. Learn "visual vocabulary"
3. Represent images by frequencies of "visual words"



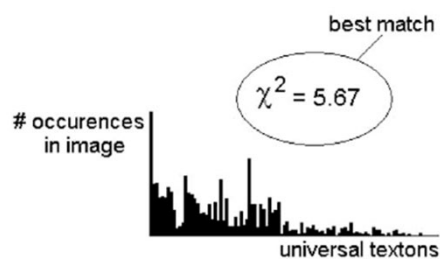
Bag of words



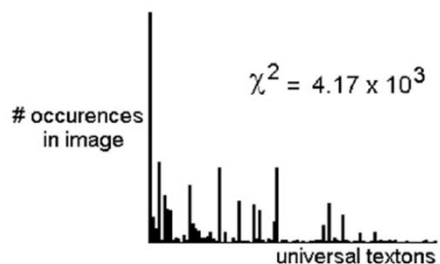
11.2 卷积核的知识



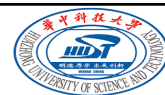
label = bedroom



label = beach



第十一讲 卷积神经网络(*Convolutional Neural Networks*)



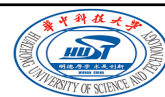
11.1 背景(*Background*)

11.2 卷积核的知识(*About Convolutional Kernels*)

11.3 卷积神经网络(*Convolutional Neural Networks*)

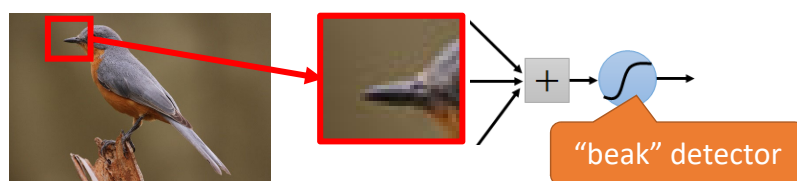
11.4 应用示例(*Case Study*)

11.3 卷积神经网络

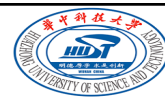


图像/视觉中采用CNN结构的原因

- 反映物体特性的模式往往只占图像中很小的一部分
 - 为了发现这些模式，神经元无需与图像中的所有像素连接
 - 神经元与图像中局部区域像素连接，所需的参数量会更少

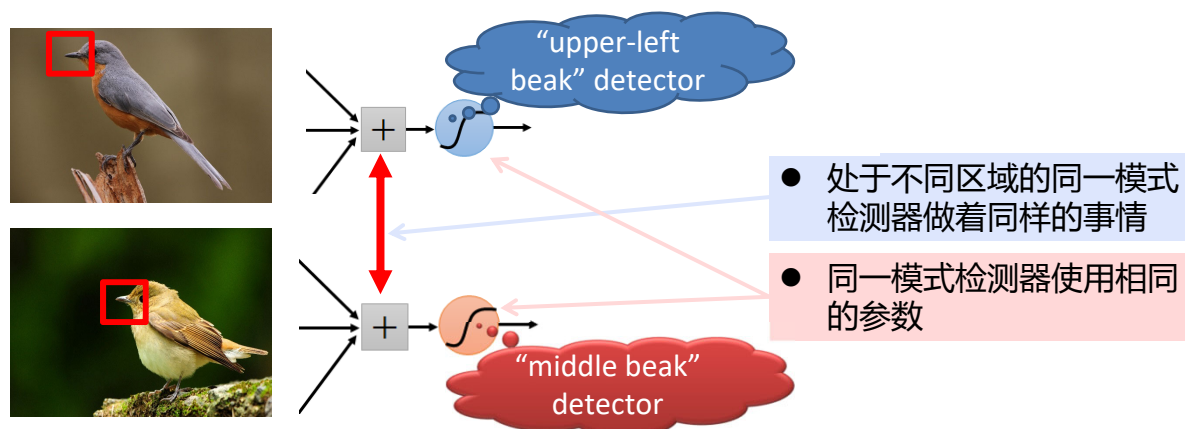


11.3 卷积神经网络

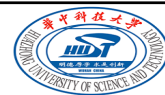


图像/视觉中采用CNN结构的原因

- 物体同样的模式可能出现在图像中的不同区域



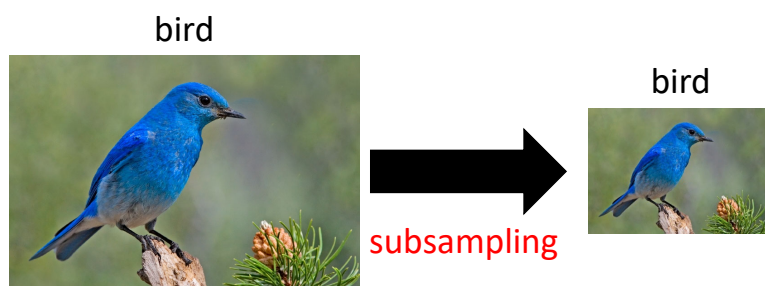
11.3 卷积神经网络



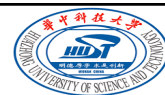
图像/视觉中采用CNN结构的原因

➤ 下采样不会改变物体的类别属性

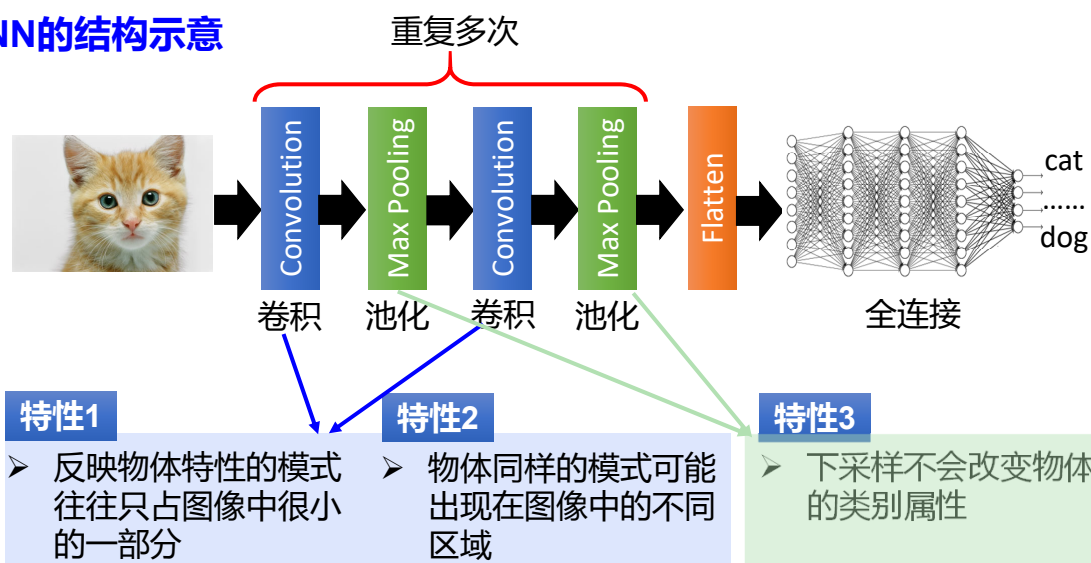
- 下采样降低了图像大小
- 网络处理较小的图像需要的参数量也随之减少



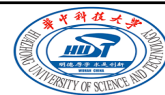
11.3 卷积神经网络



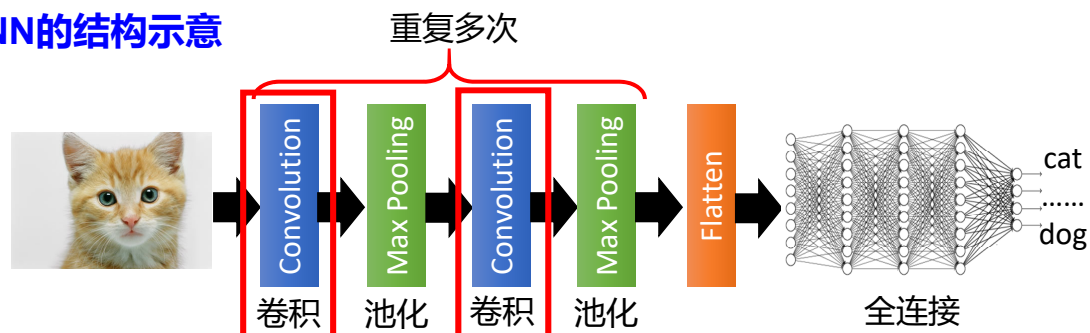
CNN的结构示意



11.3 卷积神经网络



CNN的结构示意



特性1

- 反映物体特性的模式往往只占图像中很小的一部分

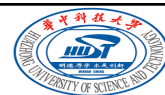
特性2

- 物体同样的模式可能出现在图像中的不同区域

特性3

- 下采样不会改变物体的类别属性

11.3 卷积神经网络



CNN—卷积层工作过程介绍

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1
Matrix

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2
Matrix

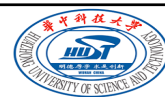
⋮

特性1

- 每个滤波器用于检测一种模式

滤波器是可通过学习获得的网络参数

11.3 卷积神经网络



CNN—卷积层工作过程介绍

stride=1

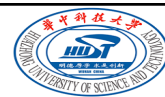
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

11.3 卷积神经网络



CNN—卷积层工作过程介绍

stride=1

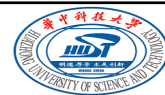
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

11.3 卷积神经网络



CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

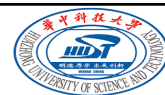
6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

3

11.3 卷积神经网络



CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

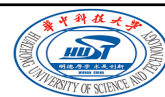
6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

3

11.3 卷积神经网络



CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

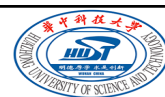
6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1



11.3 卷积神经网络



CNN—卷积层工作过程介绍

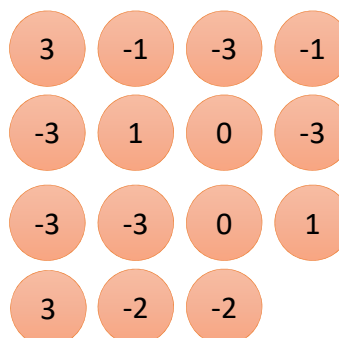
stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

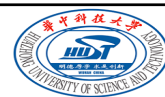
6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1



11.3 卷积神经网络



CNN—卷积层工作过程介绍

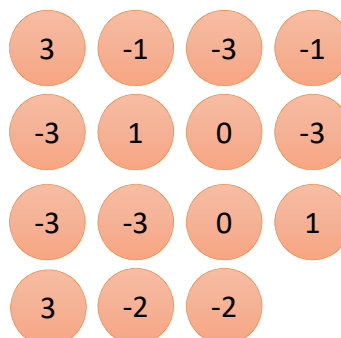
stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

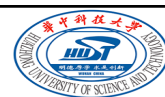
6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1



11.3 卷积神经网络



CNN—卷积层工作过程介绍

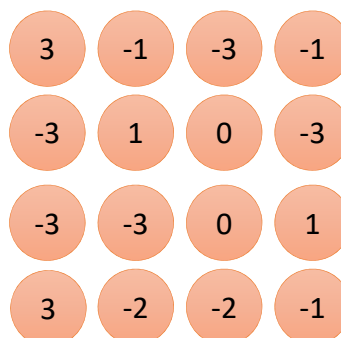
stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

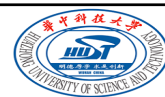
6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1



11.3 卷积神经网络



CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

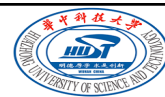
Filter 1

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

特性2

- 相同的模式处于图像的不同位置，可以用参数相同的检测器

11.3 卷积神经网络



CNN—卷积层工作过程介绍

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

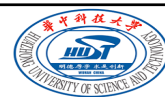
所有滤波器做同样的操作

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

11.3 卷积神经网络



CNN—卷积层工作过程介绍

stride=1

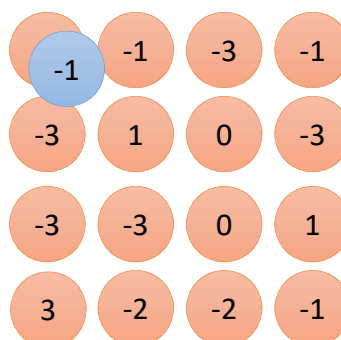
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

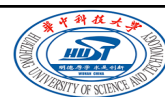
所有滤波器做
同样的操作

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2



11.3 卷积神经网络



CNN—卷积层工作过程介绍

stride=1

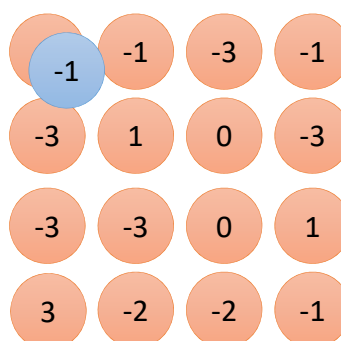
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

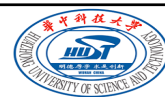
所有滤波器做
同样的操作

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2



11.3 卷积神经网络



CNN—卷积层工作过程介绍

stride=1

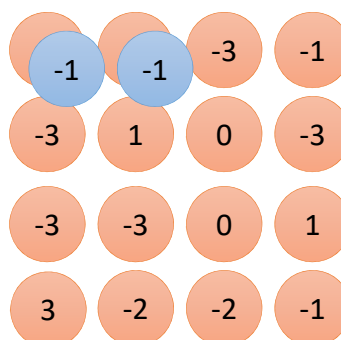
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

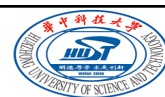
所有滤波器做
同样的操作

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2



11.3 卷积神经网络



CNN—卷积层工作过程介绍

stride=1

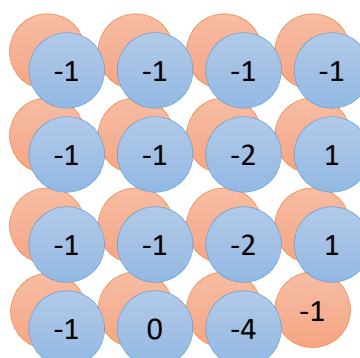
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

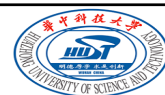
所有滤波器做
同样的操作

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2



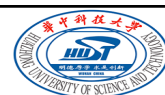
11.3 卷积神经网络



CNN—卷积层工作过程介绍



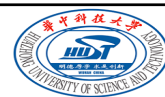
11.3 卷积神经网络



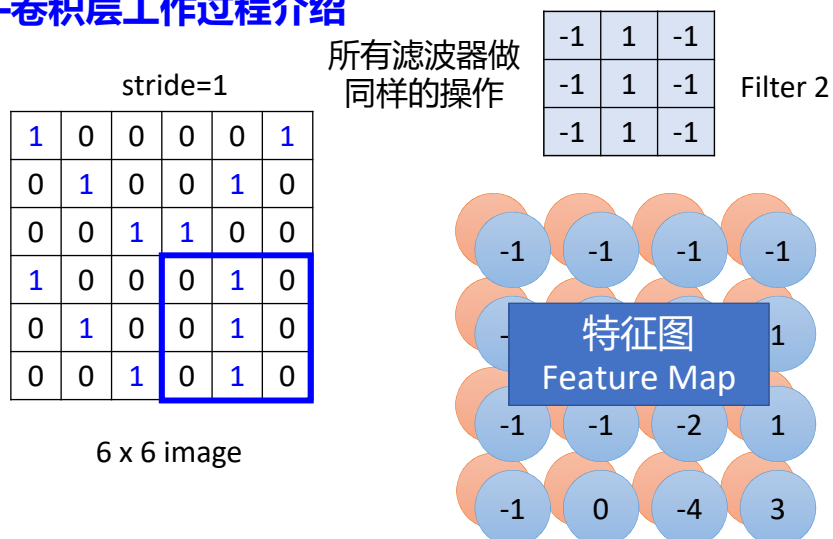
CNN—卷积层工作过程介绍



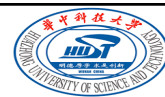
11.3 卷积神经网络



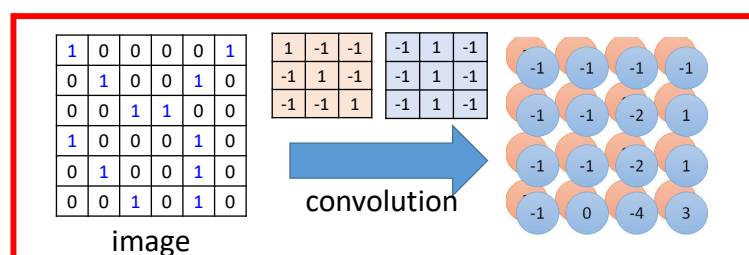
CNN—卷积层工作过程介绍



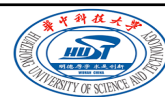
11.3 卷积神经网络



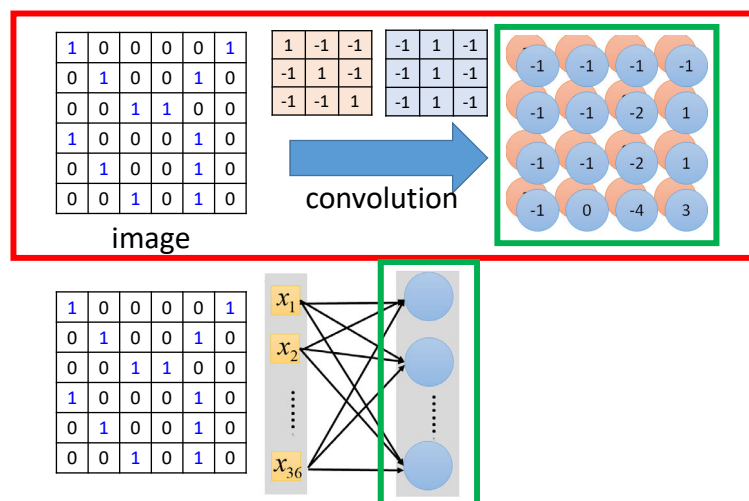
CNN—卷积层



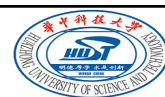
11.3 卷积神经网络



CNN—卷积层与全连接层



11.3 卷积神经网络



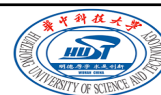
1	-1	-1
-1	1	-1
-1	-1	1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

- 1: 1
- 2: 0
- 3: 0
- 4: 0
- ⋮
- 7: 0
- 8: 1
- 9: 0
- 10: 0
- ⋮
- 13: 0
- 14: 0
- 15: 1
- 16: 1

11.3 卷积神经网络



1	-1	-1
-1	1	-1
-1	-1	1

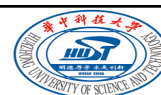
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

1: 1
2: 0
3: 0
4: 0
:
7: 0
8: 1
9: 0
10: 0
:
13: 0
14: 0
15: 1
16: 1

11.3 卷积神经网络



1	-1	-1
-1	1	-1
-1	-1	1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

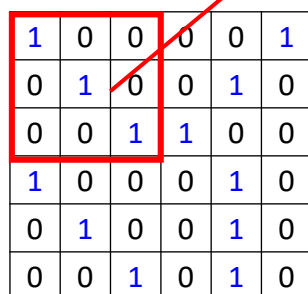
6 x 6 image

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

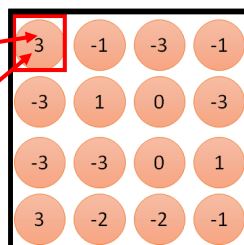
1: 1
2: 0
3: 0
4: 0
:
7: 0
8: 1
9: 0
10: 0
:
13: 0
14: 0
15: 1
16: 1

3

11.3 卷积神经网络

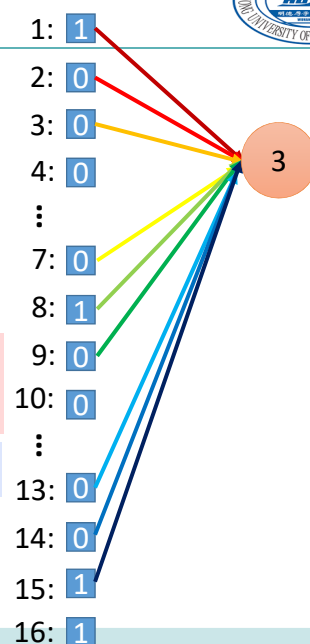


6 x 6 image

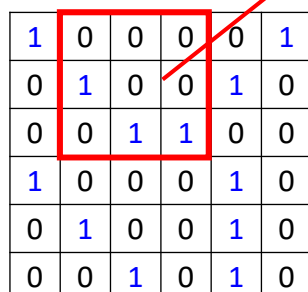


仅仅需要9个参数,
而不是全连接

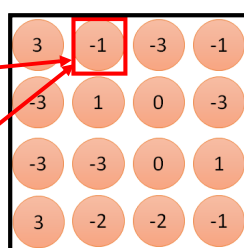
参数量减少



11.3 卷积神经网络



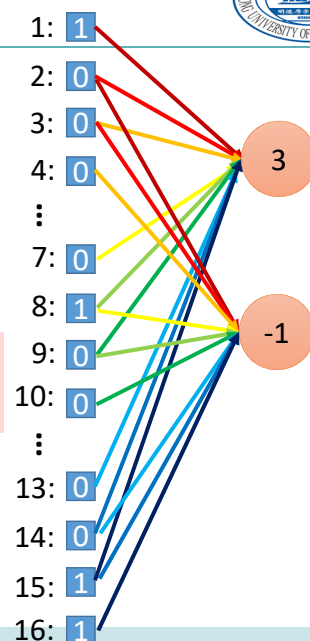
6 x 6 image



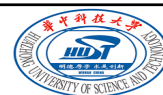
仅仅需要9个参数,
而不是全连接

参数量减少

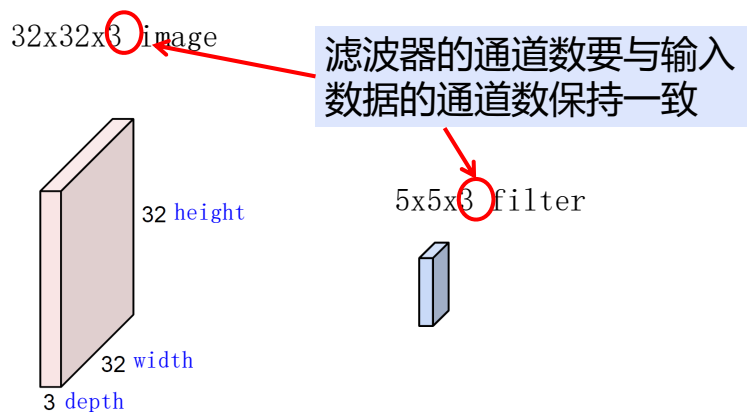
甚至可以更少



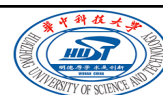
11.3 卷积神经网络



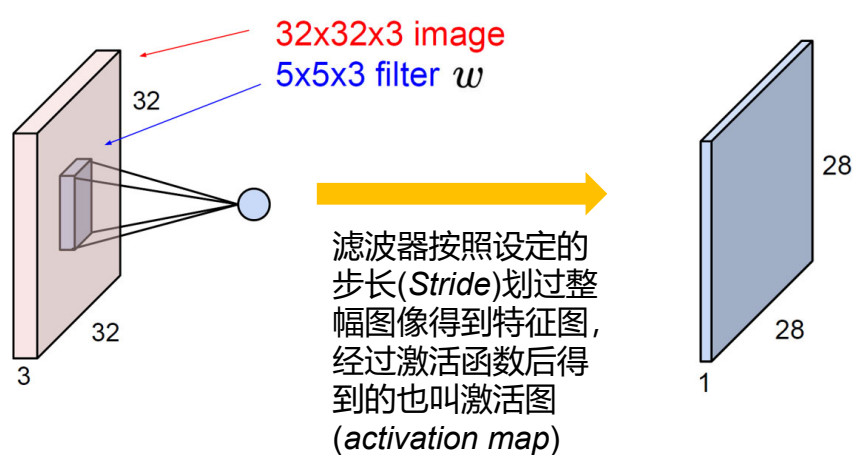
彩色图像的卷积层



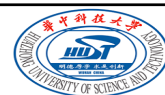
11.3 卷积神经网络



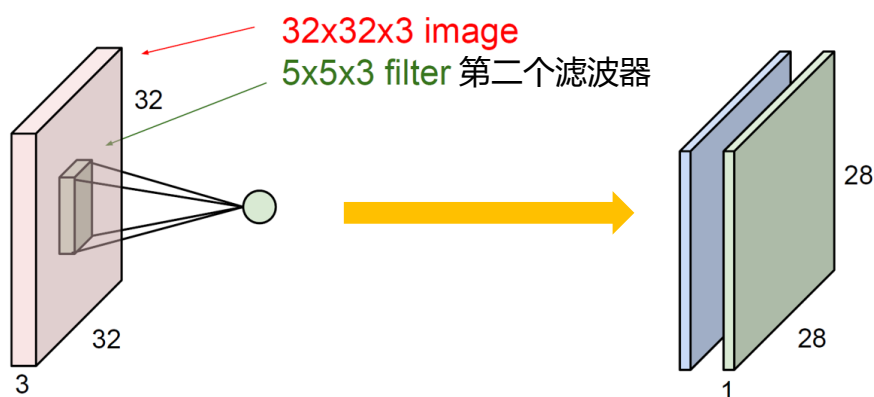
彩色图像的卷积层



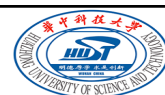
11.3 卷积神经网络



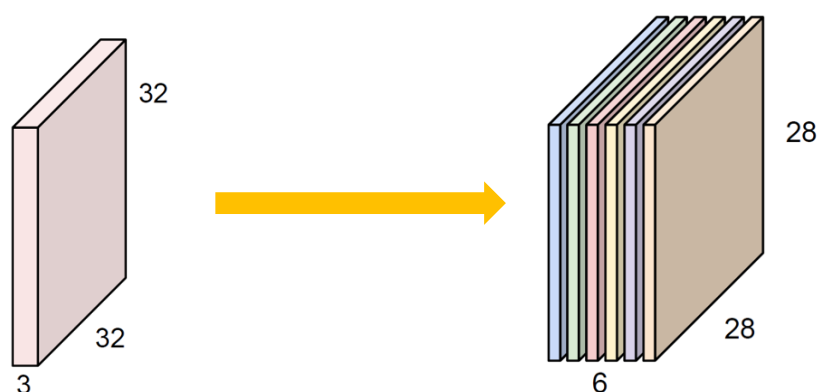
彩色图像的卷积层



11.3 卷积神经网络

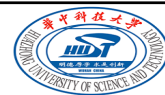


彩色图像的卷积层

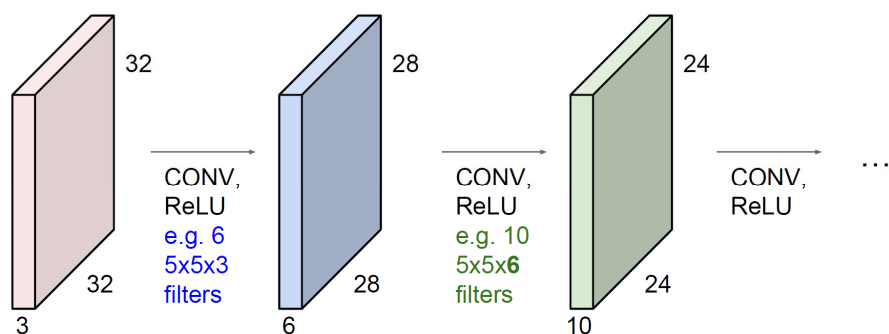


- 如果有6个5*5*3的滤波器，就可以分别得到6个特征图(激活图)
- 6个特征图堆叠在一起就构成了一幅28*28*6的“新图像”

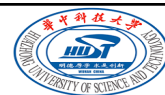
11.3 卷积神经网络



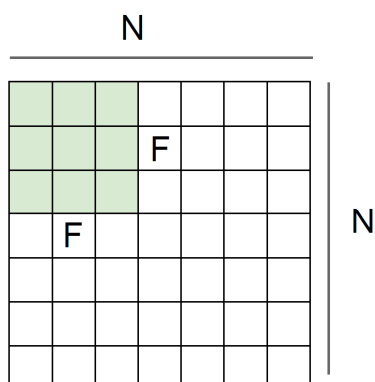
彩色图像的一系列卷积层操作



11.3 卷积神经网络



步长(Stride)、卷积后的图像大小



Output size:
 $(N - F) / \text{stride} + 1$

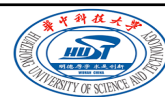
e.g. $N = 7, F = 3$:

stride 1 $\Rightarrow (7 - 3) / 1 + 1 = 5$

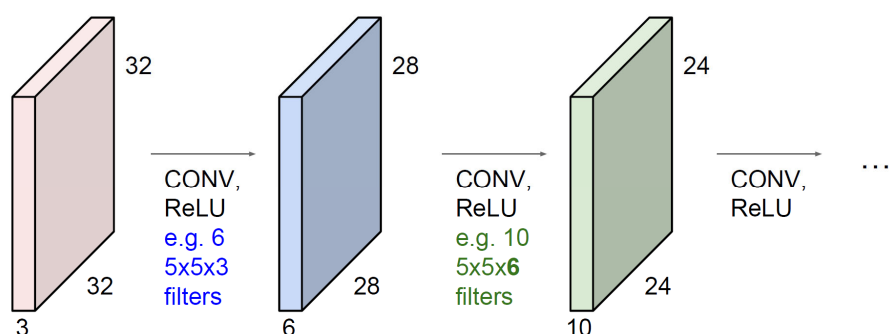
stride 2 $\Rightarrow (7 - 3) / 2 + 1 = 3$

~~stride 3 $\Rightarrow (7 - 3) / 3 + 1 = 2.33$~~

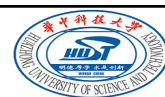
11.3 卷积神经网络



步长(Stride)、卷积后的图像大小



11.3 卷积神经网络



填充(Padding)以保持卷积后的图像大小不变

0	0	0	0	0	0			
0								
0								
0								
0								

- 实际应用中，填充常常就是在边界处补0

Input: 7*7; Filter: 3*3; Stride: 1; Pad: 1

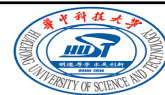
Output: 7*7; $((N-F)/S+1)$

- 保证输出与输入尺寸一样时，确定填充值的一般表达式：

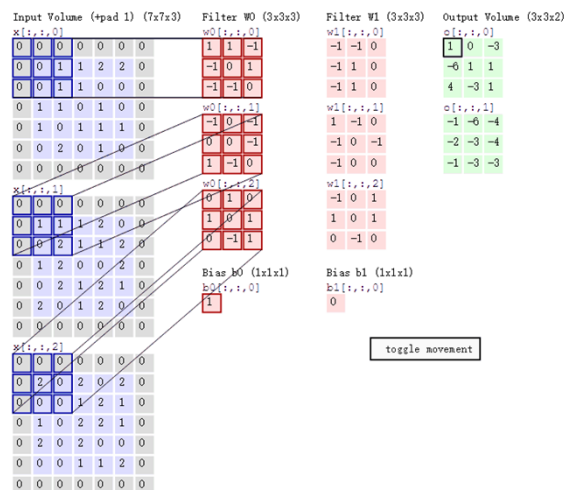
Input: $N*N$; Filter: $F*F$; Stride: S ;

Pad: $P=(N(S-1)+F-S)/2$

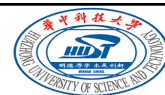
11.3 卷积神经网络



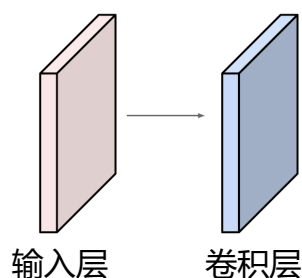
彩色图像、步长为2的卷积过程：



11.3 卷积神经网络



卷积层的模型参数数量计算示例：



输入层： 图像大小为32*32*3

卷积计算： 10个5*5*3的滤波器，Stride为1，Pad为2

卷积层： 卷积后得到的特征图大小是多少

1个滤波器： $(32+2*2-5)/1+1 = 32$

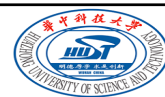
10个滤波器： $32*32*10$

连接输入与卷积层的参数量是多少

1个滤波器： $5*5*3+1 = 76$

10个滤波器： $10*76 = 760$

11.3 卷积神经网络



卷积层的模型参数量计算总结:

输入大小为 $W_1 \times H_1 \times D_1$, 滤波器个数为 K , 尺寸为 F , $Stride$ 为 S , 零填充为 P

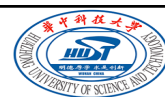
卷积后得到的特征图大小是 $W_2 \times H_2 \times D_2$, 这里:

$$W_2 = \frac{W_1 - F + 2P}{S} + 1, \quad H_2 = \frac{H_1 - F + 2P}{S} + 1, \quad D_2 = K$$

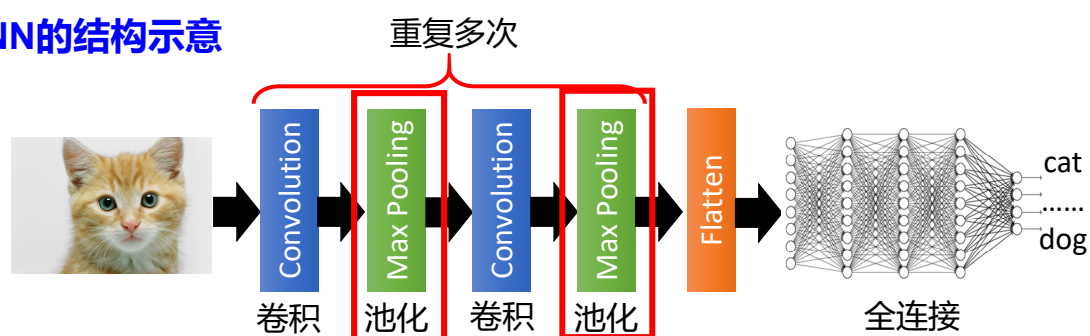
这一层的参数量是:

$$(F \cdot F \cdot D_1) \cdot K + K$$

11.3 卷积神经网络



CNN的结构示意



特性1

- 反映物体特性的模式往往只占图像中很小的一部分

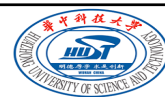
特性2

- 物体同样的模式可能出现在图像中的不同区域

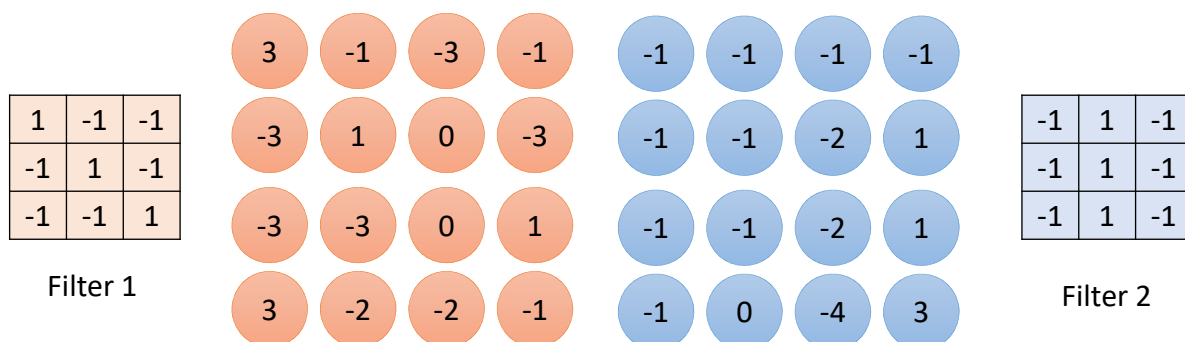
特性3

- 下采样不会改变物体的类别属性

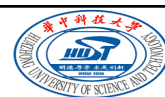
11.3 卷积神经网络



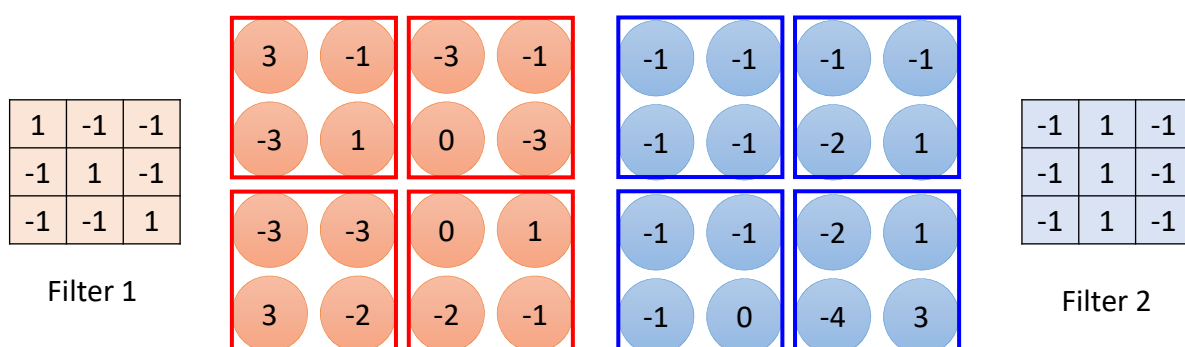
CNN – Max Pooling



11.3 卷积神经网络

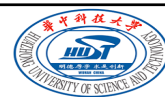


CNN – Max Pooling

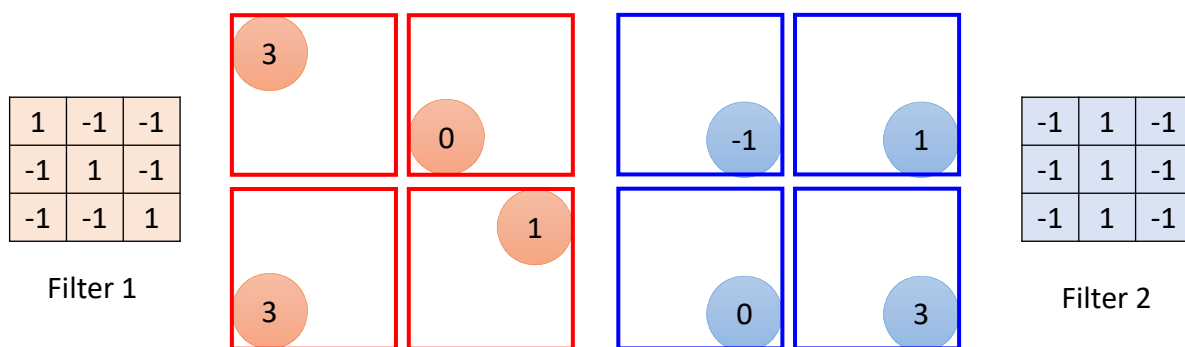


按照步长为2, 2×2 的尺寸做池化(Pooling)

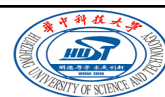
11.3 卷积神经网络



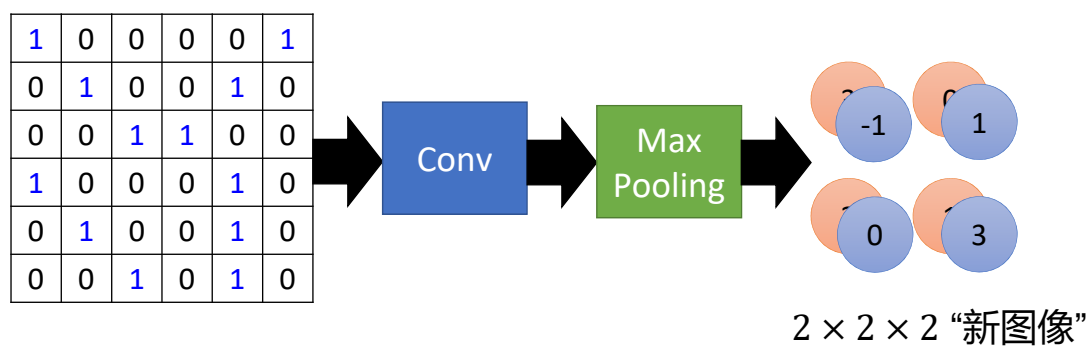
CNN – Max Pooling



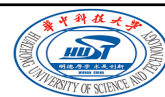
11.3 卷积神经网络



CNN – Max Pooling



11.3 卷积神经网络



池化层的模型参数量计算总结:

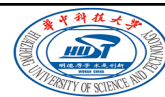
输入大小为 $W_1 \times H_1 \times D_1$, 池化(Pooling)窗口的尺寸为 F , 步长(Stride)为 S

池化后得到的特征图大小是 $W_2 \times H_2 \times D_2$, 这里:

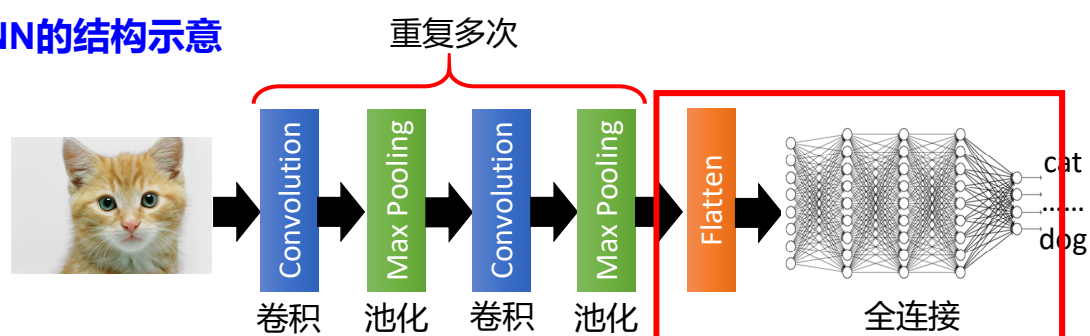
$$W_2 = \frac{W_1 - F}{S} + 1, \quad H_2 = \frac{H_1 - F}{S} + 1, \quad D_2 = D_1$$

池化层的参数量是: 0

11.3 卷积神经网络



CNN的结构示意



特性1

- 反映物体特性的模式往往只占图像中很小的一部分

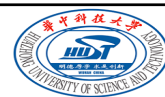
特性2

- 物体同样的模式可能出现在图像中的不同区域

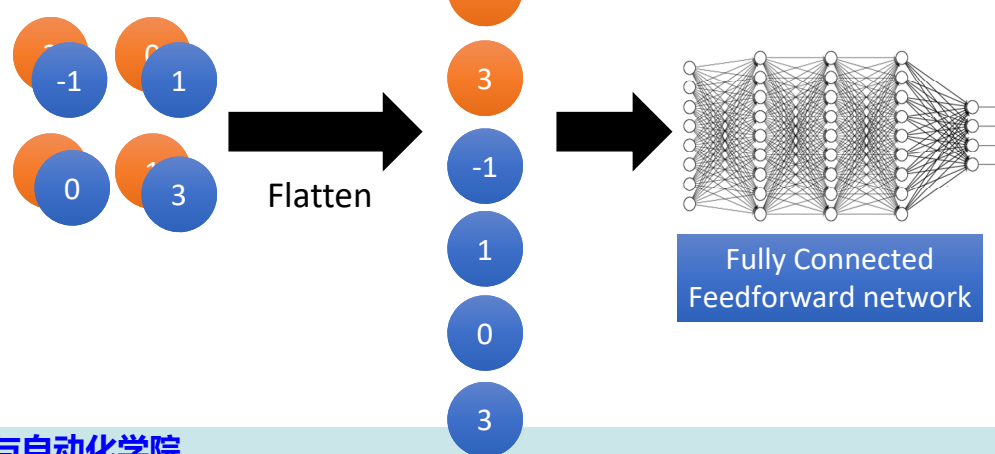
特性3

- 下采样不会改变物体的类别属性

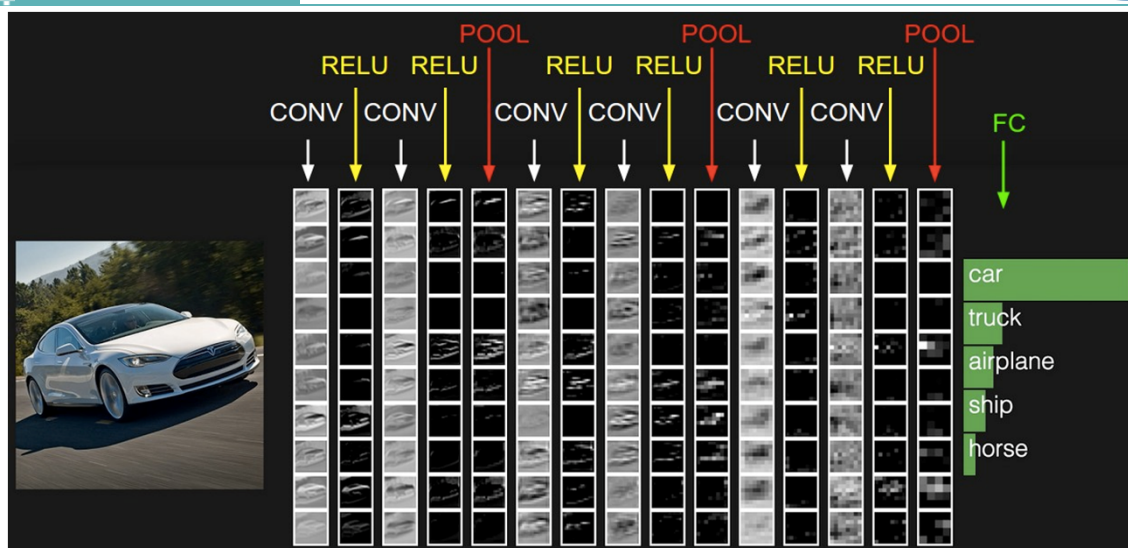
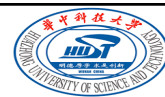
11.3 卷积神经网络



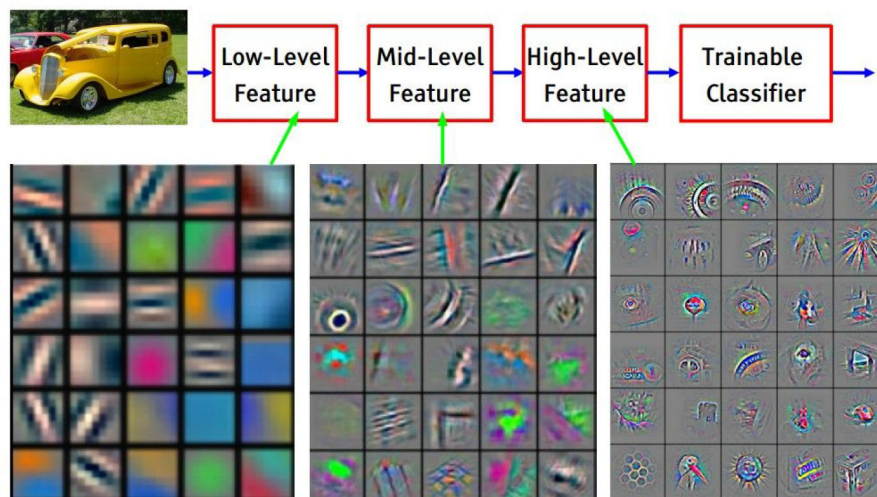
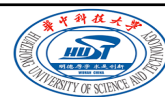
CNN – Flatten



11.3 卷积神经网络



11.3 卷积神经网络

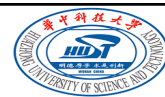


所有参数都是
通过学习得到

端到端的学习

Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

第十一讲 卷积神经网络(*Convolutional Neural Networks*)



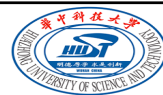
11.1 背景(*Background*)

11.2 卷积核的知识(*About Convolutional Kernels*)

11.3 卷积神经网络(*Convolutional Neural Networks*)

11.4 应用示例(*Case Study*)

第十一讲 卷积神经网络(*Convolutional Neural Networks*)



11.1 背景(Background)

CNN在计算机视觉影响力的一段往事

11.2 卷积核的知识(About Convolutional Kernels)

滤波器组、纹理基元、视觉词典

11.3 卷积神经网络(Convolutional Neural Networks)

卷积层、池化层、全连接层。。。

11.4 应用示例(Case Study)

图像识别应用